

Sprint Action Plan: Person 4: Integration Lead

Qwen2.5-0.5B Monkey-Patching, Prefill Throughput & Model Forward Pass • Adaptive Attention Computation Sprint

Assigned Role: Person 4: Integration Lead	Estimated Workload: 9.5 Hours	Sprint Timeline: 0.0h – 9.5h
Core Discipline: Deep Learning Frameworks / Pretrained Model Engineering	Primary Handoff To: Person 6 (Synthesis Lead)	Target Hardware: Dual T4 (16GB)

1. Role Mission & Strategic Objective

Integrate custom linear and adaptive attention kernels into a pretrained Transformer model (Qwen/Qwen2.5-0.5B). Validate end-to-end forward passes, benchmark prefill latency on Kaggle T4, and ensure memory footprints remain within 16GB.

2. Input Dependencies & Final Deliverables

Type	Item / Specification	Source / Destination	Deadline / Gate
INPUT	Qwen/Qwen2.5-0.5B Hugging Face weights	Hugging Face Hub	Hour 0.0
INPUT	Mock kernel interface	Person 1 (Theory Lead)	Hour 1.0
INPUT	Production adaptive attention kernel	Person 1 (Theory Lead)	Hour 7.5
OUTPUT	Model monkey-patch script (patch_qwen.py)	Team Repository	Hour 7.5
OUTPUT	Prefill latency benchmark logs (qwen_prefill.csv)	Person 6 (Synthesis Lead)	Hour 8.0
OUTPUT	Patched model forward pass verification logs	Team Repository	Hour 8.5

3. Technical Specifications & Contract Signature

Key Code / Mathematical Contract for Person 4: Integration Lead:
- Model Baseline: Qwen/Qwen2.5-0.5B loaded in torch.float16 (~1.0 GB baseline VRAM footprint) - Target Module: Monkey-patch Qwen2Attention.forward() in all transformer layers - I/O Interface: Accept hidden_states (B, N, D), attention_mask, position_ids, past_key_values - Sweep Sequences: Benchmark prefill latency across prompt lengths $N \in \{1024, 2048, 4096, 8192\}$

4. Hour-by-Hour Master Action Checklist — Person 4: Integration Lead

Window	Specific Actions & Step-by-Step Instructions	Deliverable / Evidence	Status
0.0 – 1.0 h	Setup Hugging Face environment on Kaggle T4; verify Qwen2.5-0.5B baseline FP16 inference.	Baseline inference verification	[] Done
1.0 – 3.0 h	Construct MockAttention monkey-patch harness to intercept and replace Qwen2Attention blocks.	patch_qwen.py (Monkey-patch infrastructure)	[] Done
3.0 – 5.5 h	Validate forward pass and tensor shapes using mock linear kernel; test generation loop stability.	Forward pass validation passing	[] Done
5.5 – 7.5 h	Profile prefill latency and peak VRAM across prompt lengths N = 1K to 8K comparing SDPA vs Linear.	qwen_prefill.csv benchmark logs	[] Done
7.5 – 8.5 h	Drop in Person 1's real AdaptiveRankAttention kernel; verify forward pass and prefill latency.	Patched model integration complete	[] Done
8.5 – 9.5 h	Push patched model codebase and verification scripts; participate in presentation rehearsal.	Repository codebase push & sign-off	[] Done

5. Operational Guardrails & Failure Protocols

Constraint / Risk	Enforcement Rule & Exact Action
Model Weight VRAM Bloat	Strictly load model in FP16 to preserve > 12 GB VRAM for long-context KV activations.
RoPE / Positional Incompatibility	Ensure rotary embeddings (RoPE) are applied to Q and K prior to linear kernel computation.
KV Cache Misalignment	Verify recurrent state accumulation matches autoregressive decoding requirements.

6. Definition of Done (DoD) & Sign-Off Checklist

[]	patch_qwen.py monkey-patches Qwen2Attention cleanly without modifying library source.
[]	End-to-end forward pass succeeds on Qwen2.5-0.5B without NaNs or CUDA errors.
[]	Prefill latency benchmarked across N = 1K to 8K.
[]	VRAM usage stays strictly under 14 GB across all benchmark runs.